A Condition-Indexed Operating Envelope and Representation-Repair Study for Automatic Modulation Classification under Structured Interference

Anonymous Author(s)

***Abstract*—Uniform model rankings hide a central design question in automatic modulation classification (AMC): under structured interference, the ranking of a compact spectral representation and a much larger I/Q-domain classifier can itself depend on physical conditions. We study this rank exchange in source-disjoint 3GPP TR 38.901 TDL-profile simulations, comparing VIMD-Net, a 39500-parameter spectral network, with strong I/Q-domain comparators across the complete SNR–SIR plane. The original sealed confirmatory family resolves a positive whole-configuration contrast of 4.57 percentage points ($[3.65, 5.49]$ percentage points) for the complete configuration over its shared spectral backbone, while the two component-level interventions (teacher form and route count) are bounded rather than individually resolved. In a post-hoc severe-interference analysis at SIR $= -15$ dB, the compact model gains 9.74 percentage points over MCLDNN and 10.46 percentage points over IQFormer-inspired, with positive paired intervals and the same direction in every seed. A two-variable occupancy–SIR envelope fitted on 32 in-distribution and hard-interference cells attains RMSE 6.20 percentage points on 81 held-out regime cells against IQFormer-inspired, reducing RMSE from 15.28 percentage points for a constant baseline ($R^2_{\text{skill}} = 0.84$); sign agreement is 0.975 against a majority-sign baseline of 0.963, so magnitude skill, not sign accuracy, is the principal validation. A failed clean-retention gate is then localized and repaired prospectively: a received-I/Q sidecar raises the total model size from 39500 to 46794 parameters (+18% parameters, about +3% MACs) while improving clean macro-F1 by 11.77 percentage points and hard-interference macro-F1 by 4.37 percentage points; these Tier-2 results are exploratory and outside the sealed family. The paper reports a simulation-only operating envelope and a repair path, not uniform superiority.**

***Index Terms*—Automatic modulation classification, structured interference, vehicular Doppler, operating envelope, condition-dependent model comparison, representation repair, compact neural networks.**

Evidence status. The original sealed family resolves one positive whole-configuration contrast (A5 vs A0); the two component contrasts and the clean-retention gate do not pass. The severe-interference, operating-envelope, and Tier-2 repair analyses are exploratory or post-hoc and are labeled as such. All conclusions are restricted to source-disjoint simulation.

I. Introduction

AUTOMATIC modulation classification supports spectrum monitoring and blind receiver configuration by inferring a waveform's modulation from received samples. Deep AMC has progressed from convolutional I/Q classifiers [1], [2] to complex-valued, multi-stream, recurrent, and Transformer-like representations [3]–[7]. Compact architectures remain attractive when inference cost matters [8], [9], but average accuracy alone does not say where a compact representation is preferable.

This omission is consequential in mobile and vehicular receivers. A high-mobility terminal facing structured interference (a sweep, tone, pulse, partial-band process, or modulated out-of-taxonomy emitter overlapping the target in time and frequency) must decide on a limited compute budget whether to run a compact spectral front end or a high-capacity I/Q path. The same model can trail a high-capacity I/Q classifier in weak or narrowband interference yet lead it in a severe, spectrally occupied corner. Averaging these regimes produces a ranking but erases the design rule.

This paper asks a narrower question: *can the sign and size of the compact-model advantage be indexed by physical condition variables, and can the boundary revealed by that index be repaired?* We answer using a frozen source-disjoint simulation campaign, artifact-only post-hoc analysis, and prospective Tier-2 experiments. The base model, VIMD-Net, maps a complex STFT to target-dominant, jammer-dominant, and unexplained-or-ambiguous routes; its supervision is physics-informed in the sense that a simulation-component teacher, available only during training, is derived from independently tracked target, jammer, and residual components. We do not claim that a specific routing or teacher component is individually confirmed (Section V-A); the contribution is the boundary and its repair, not the architecture itself.

The contributions are:

- a condition-indexed rank exchange, exposed over the complete structured-interference SNR–SIR plane, between a compact spectral model and stronger I/Q-domain classifiers, instead of a single marginal ranking;
- a low-dimensional occupancy–SIR empirical envelope whose magnitude skill is evaluated on regime cells excluded from fitting, with an explicit trivial-baseline comparison and a bounded, non-causal interpretation;
- a separation of the original sealed whole-configuration contrast from post-hoc severe-interference evidence, using paired statistics, multiplicity control, and full-region visualization so a favorable corner is not read as uniform superiority; and
- a prospective diagnosis-and-repair chain that turns a failed clean-retention gate into a representation-information finding: a received-I/Q sidecar increases the total model size only modestly while repairing clean and hard-interference performance, and a spectral capacity ladder shows that width alone does not remove the boundary.

The claim is deliberately not that VIMD-Net is uniformly best. It is that the compact and high-capacity representations exchange rank in a reproducible, condition-dependent way, that this boundary can be described by physical covariates, and that its principal clean boundary is a repairable representation gap rather than an unavoidable cost of compactness.

II. Related Work and Claim Boundary

A. Strong and Efficient AMC Representations

CNN and recurrent AMC systems learn directly from I/Q sequences [1], [2], [5]; later work preserves complex structure, fuses temporal and spatial streams, or applies attention [3], [4], [6], [7]. Compact and efficient AMC has received sustained attention for cost-sensitive deployment [8], [9]. These lines of work motivate strong I/Q-domain comparators and prevent us from presenting a lightweight architecture or a fusion scheme as the primary novelty.

B. Robust and Cross-Domain AMC

Channel adaptation, masked or contrastive learning, and interference-tolerant features address different nuisance sources [10]–[13]. Overlapped-signal classification, denoising masked autoencoding, and cross-domain prototypical adaptation further populate the 2024–2026 frontier [14]–[16]. These works establish that "robust AMC" and "cross-domain AMC" are mature themes; we therefore do not claim robustness or adaptation as novelty.

C. Interference-Aware and Malicious-Interference AMC

Interference-tolerant recognition [13] and tensor-based malicious-interference recognition for MIMO [17] already address structured interference directly, including low-SIR conditions. Our setting differs: we do not perform source reconstruction or tensor separation, and we do not aim to be a universally interference-robust classifier. We study how the *ranking* of compact spectral versus high-capacity I/Q models changes with the interference condition itself.

D. Distinction: Condition-Indexed Ranking and Repair

Existing work has made substantial progress in interference-tolerant recognition, domain adaptation, lightweight architectures, and multi-representation fusion (including recent knowledge-guided and multimodal preprints [18]–[21]). Our contribution is not another claim that a particular representation dominates on average. We study the condition dependence of the ranking itself, fit a low-dimensional empirical boundary to that rank exchange, expose the negative regions rather than reporting only favorable averages, and prospectively intervene on the revealed representation boundary. To our knowledge, prior AMC studies have largely optimized or compared marginal performance rather than explicitly modeling the condition-dependent rank exchange between compact spectral and high-capacity I/Q representations and then intervening on the boundary; we state this as a positioning claim, not an exhaustive systematic-review result.

III. System Model and VIMD Configuration

A. Mobile Received Signal and Paired Views

We model a receiver-side AMC front end under terrestrial high-mobility conditions. For source sequence $s[n]$ and jammer $u[n]$, one received window is

$$x[n] = \mathcal{R}(\mathcal{H}_s(s)[n] + \gamma\mathcal{H}_u(u)[n] + w[n])\,, \tag{1}$$

where $\mathcal{H}_s$ and $\mathcal{H}_u$ are independently drawn fading and mobility operators (parameterized by carrier frequency, speed range, and 3GPP TR 38.901 TDL-A/C/D profiles during training, with TDL-B/E held out for channel evaluation), γ controls SIR, w is receiver noise, and $\mathcal{R}$ applies the frozen receiver chain. Training pairs two independently impaired observations of the same immutable source; inference uses one view. Pairing is therefore a training and statistical-alignment device, not additional test-time information. This model is a controlled simulation of a mobile receiver, not a complete V2X geometry; no MAC- or network-layer V2X performance is claimed.

Unanchored in-taxonomy cochannel collisions are excluded: without a physical target anchor, exchanging two admissible emitters can preserve the mixture while changing the requested single label, which requires a different label definition.

B. Three-Route Spectral Allocation

Let $Z = \mathrm{STFT}(x) \in \mathbb{C}^{F\times T}$. The front end receives real, imaginary, and log-magnitude planes and produces logits A_s, A_j, A_o on the same lattice. The allocation masks are

$$\begin{aligned} &[M_s, M_j, M_o] = \mathrm{softmax}([A_s, A_j, A_o]),\\ &M_s + M_j + M_o = 1. \end{aligned} \tag{2}$$

An environment encoder produces a scalar ambiguous-route gate λ and a bounded bypass ρ. The modulation and jammer routes are

$$Z_m = (M_s + \lambda M_o + \rho) \odot Z, \tag{3}$$

$$Z_j = (M_j + (1-\lambda) M_o) \odot Z. \tag{4}$$

Because these weights are real and nonnegative, every retained coefficient keeps the mixture STFT phase. This elementary property does not imply source identifiability.

C. Simulation-Component Teacher

During training only, independently tracked post-channel target, jammer, and residual components define powers P_s, P_j, P_u on the student's lattice. With $q_k = P_k/(P_s + P_j + P_u)$, the fixed teacher is

$$M_s^\star = [q_s - q_j]_+, \qquad M_j^\star = [q_j - q_s]_+, \tag{5}$$

$$M_o^\star = q_u + 2\min(q_s, q_j). \tag{6}$$

The routes are nonnegative and sum to one; negligible-energy cells are assigned to the uncertainty route by the finite-precision implementation. We call this construction *physics-informed* because the supervision uses simulator-tracked components; it does not imply that the margin teacher or the three-route allocation is individually confirmed (Section V-A). The teacher is not needed at inference.

TABLE I
INFORMATION CONTRACT ACROSS STAGES.

Quantity	Training	Test inference	Post-hoc analysis
Received mixture	Yes	Yes	Yes
Tracked target component	Teacher only	No	Simulator audit
Tracked jammer component	Teacher only	No	Occupancy estimate
Clean counterfactual	No	No	Tier-2 probe only
True SIR	Simulator metadata	No	Envelope covariate

TABLE II
NOTATION AND MODEL IDENTIFIERS.

Identifier	Meaning
A0	direct spectral reference backbone
A5 / VIMD-Net	sealed full configuration
A7	residual-free control variant
S / M / L	prospective spectral capacity tiers
sidecar	Tier-2 received-I/Q repair model
CSSL	adaptation-oriented reference [12], [22]
IQFormer-inspired	unified-retraining strong I/Q reference [7]
macro-F1	unweighted mean of class-wise F1

D. Training Objective

The total loss is

$$\mathcal{L} = \mathcal{L}_{\mathrm{mod}} + \alpha\mathcal{L}_{\mathrm{teacher}} + \beta\mathcal{L}_{\mathrm{jammer}} + \gamma\mathcal{L}_{\mathrm{link}} + \eta\mathcal{L}_{\mathrm{contrast}}, \quad (7)$$

where $\mathcal{L}_{\mathrm{mod}}$ is the modulation classification loss, $\mathcal{L}_{\mathrm{teacher}}$ is a Jensen–Shannon alignment between student and teacher masks, $\mathcal{L}_{\mathrm{jammer}}$ and $\mathcal{L}_{\mathrm{link}}$ constrain the jammer-family and link-quality auxiliary objectives, $\mathcal{L}_{\mathrm{contrast}}$ is an exact-source contrastive loss that encourages condition invariance, and $\alpha, \beta, \gamma, \eta$ are nonnegative weights. The exact-source contrastive loss is an auxiliary training objective, not a headline contribution.

E. Information Contract and Complexity

Table I states which quantities are available at each stage, to remove any oracle-information ambiguity. VIMD-Net uses 39500 parameters and 41.8 million reported MACs. Table II fixes the model identifiers used throughout.

IV. EVIDENCE PROTOCOL

A. Frozen Source-Disjoint Campaign

The artifact-locked campaign contains 12 model variants, 10 algorithm seeds, 120 fits, 11 evaluation splits, and 100000 training source sequences. Target classes cover PSK, QAM, continuous-phase, and FSK families. Structured jammer families span tone, multitone, chirp, partial band, pulse, broadband, and out-of-taxonomy modulated sources. TDL-A/C/D profiles are seen during training, while TDL-B/E profiles support held-channel evaluation. Jammer family, speed, channel, combined OOD, receiver stress, and clean-retention splits share source identities across model comparisons. The receiver-stress split (ADC quantization and per-emitter synchronization) is retained in the artifact package and evaluated for the Tier-2 sidecar, but it does not enter the sealed or envelope claims of this paper.

TABLE III
COMPARATOR AND COMPLEXITY SUMMARY (HARD-INTERFERENCE SETTING).

Model	Domain	Params	MACs	Seeds
A0	spectral	8458	6.3 M	10
A5 / VIMD-Net	spectral allocation	39500	41.8 M	10
sidecar	spectral + received I/Q	46794	43.1 M	10
MCLDNN	I/Q	406070	398.2 M	10
IQFormer-inspired	multimodal I/Q	354984	355.6 M	10
CSSL	I/Q (adaptation)	8631948	155.3 M	10

TABLE IV
WHAT DIFFERS BETWEEN A0 AND A5.

Property	A0	A5
Parameter count	8458	39500
Three-route allocation	No	Yes
Simulation-component teacher	No	Yes
Multi-task objectives	No	Yes
Exact-source contrastive	No	Yes
Residual bypass	No	Yes

B. Comparator Fairness

Table III summarizes the comparators. All models are retrained under the same source-disjoint protocol; “IQFormer-inspired” and “MCLDNN” denote architecture-faithful unified-retraining references, not exact reproductions of every original training recipe. The shortened axis label “IQFormer” in the figures refers to IQFormer-inspired.

C. Original Sealed Confirmatory Family

The sealed family contains three hard-interference contrasts: the complete configuration versus A0, the margin teacher versus a proportional-power teacher, and the tri-route model versus a dual-route control. Table IV records what differs between A0 and A5 so that the whole-configuration contrast is not over-interpreted. The A5–A0 contrast is interpreted as a whole-configuration effect, not as an isolated causal effect of the teacher or route count, because it also changes capacity (about $4.7\times$ parameters) and bundles the multi-task and contrastive terms with the allocation.

D. Exploratory and Tier-2 Evidence Classes

All SIR-cell, occupancy, fusion, condition-transfer, probe, repair, and capacity analyses are exploratory. In particular, the severe SIR stratification is post hoc. Tier-2 experiments were designed after inspecting the frozen evidence and are outside the frozen confirmatory family; they cannot retroactively change a sealed conclusion. The clean-retention gate was preregistered separately and did not pass; clean behavior is therefore treated as a boundary to explain and repair, not as a preserved property of the sealed model.

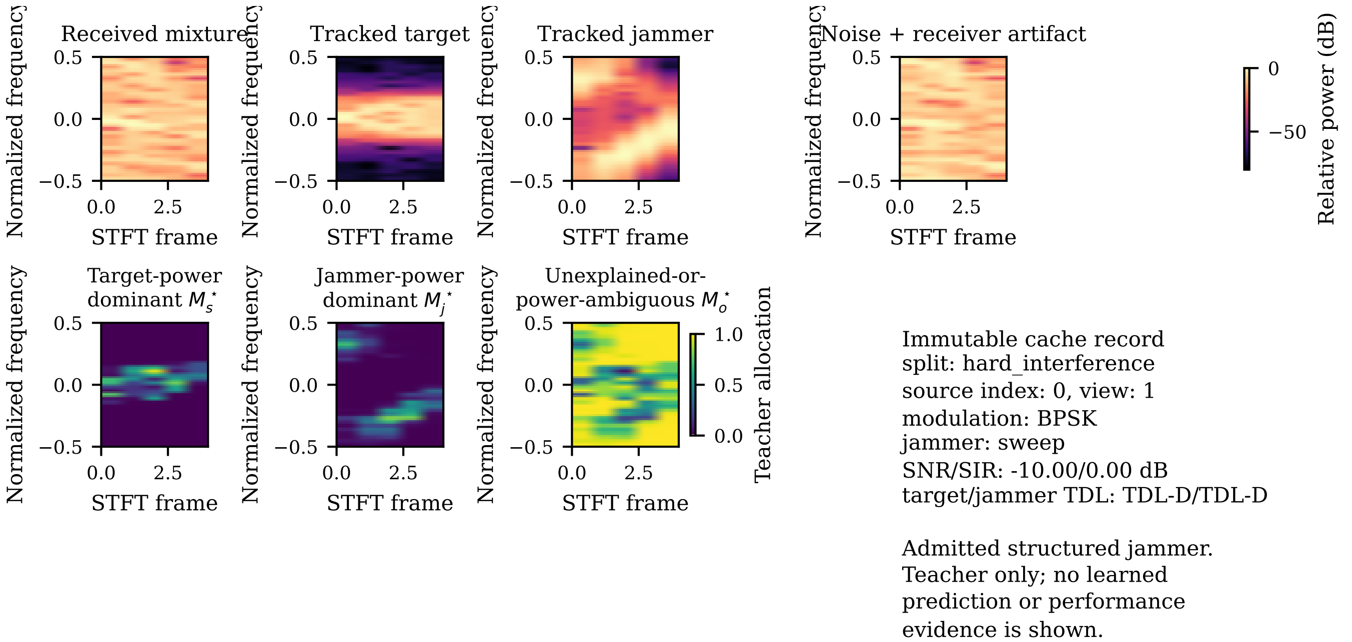

Fig. 1. A fixed-teacher proxy visualization from an immutable simulation record. The upper row shows the received mixture and tracked components; the lower row shows the three allocations on the same STFT lattice. The image explains the teacher definition and is not performance evidence.

E. Statistical Procedures

The sealed family uses a joint non-studentized max-absolute-deviation hierarchical paired bootstrap with 10000 draws, resampling class-stratified source clusters while keeping algorithm seeds as fixed blocks, so pairing is preserved at the source level and simultaneous intervals are computed from the joint max absolute deviation. The design resolution of 1.31 percentage points is the simultaneous minimum detectable effect at 80% power under this joint interval, and it is an exploratory calibration rather than a sealed significance threshold. In the severe corner, all 10 seeds share the same sign; the sign-flip permutation probability 0.0018 is a permutation test over the source-paired samples (an exact two-sided seed-level sign test would give $2/1024 \approx 0.0020$). The SIR $= -15$ dB aggregate is computed from source-paired predictions pooled across the eight SNR conditions before macro-F1 aggregation; it is therefore not the arithmetic mean of the eight displayed cell-level differences in Fig. 2.

V. Results

A. Positive Sealed Whole-Configuration Contrast and Bounded Component Effects

Table V reports the complete sealed family. The complete configuration improves over its shared A0 backbone by 4.57 percentage points, with a simultaneous interval of $[3.65, 5.49]$ percentage points. This effect is about 3.5 times the design resolution of 1.31 percentage points. The teacher-form and route-count interventions do not clear zero individually; under the same joint interval their contributions are bounded within 1.36 percentage points. Thus the configuration-level effect is resolved but cannot be assigned to either isolated component at this resolution, and because the family rule required all three contrasts, the family gate did not pass. At the same parameter tier, the residual-free A7 variant is nominally above A5 by 0.21 percentage points in the marginal hard-interference aggregate (Fig. 5), well below the design resolution and consistent with the bounded component-level result; we therefore attribute the compact frontier position to the VIMD family at this budget, not uniquely to any single component.

TABLE V
Sealed Confirmatory Hard-Interference Family. Differences Are Candidate Minus Reference in Macro-F1 Percentage Points.

Contrast	Diff.	Sim. low	Sim. high
Full configuration vs A0	4.57	3.65	5.49
Margin vs proportional teacher	0.16	-0.76	1.08
Tri-route vs dual-route	0.44	-0.47	1.36

B. Rank Exchange across the SNR–SIR Plane

Marginal hard-interference macro-F1 is 0.4406 for A5, 0.4599 for MCLDNN, and 0.4872 for IQFormer-inspired. Those averages favor the larger models, but Fig. 2 shows why no single ranking is adequate: rank changes with SNR and SIR, with A5 leading only in the severe, spectrally occupied corner.

At SIR $= -15$ dB, in a post-hoc analysis, A5 gains 9.74 percentage points over MCLDNN ($[7.52, 12.14]$) and 10.46 percentage points over IQFormer-inspired ($[6.78, 14.29]$). All 10 seeds have the same sign, and the sign-flip permutation probability is 0.0018. Of 64 inspected cells, 12 survive Holm correction; all are on the severe SIR row. These checks reinforce the corner result without converting the post-hoc

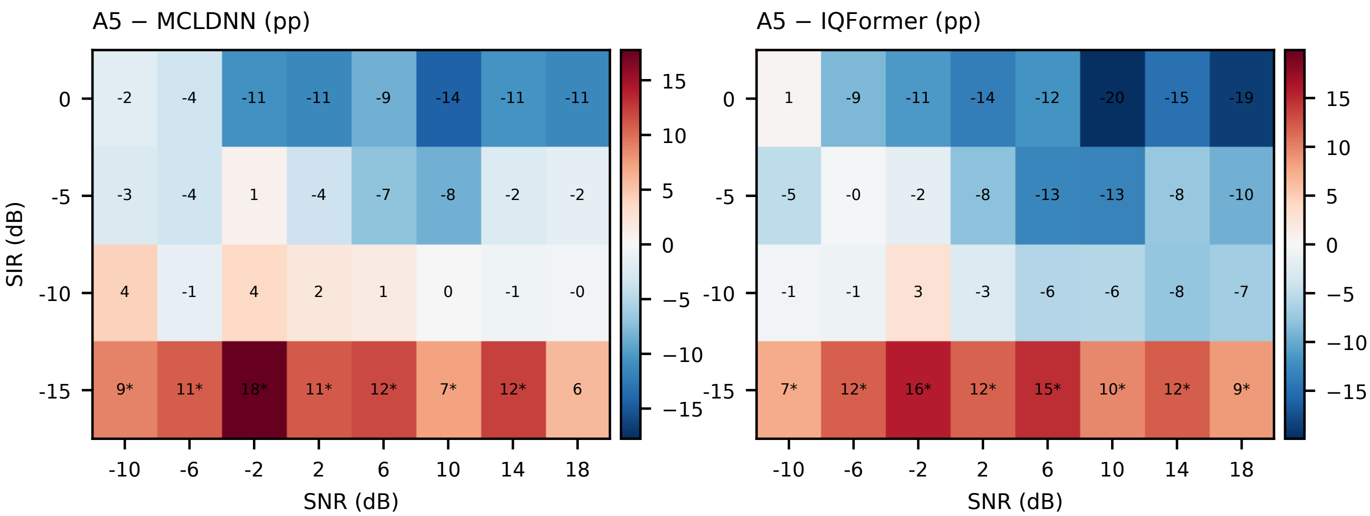

Fig. 2. Post-hoc operating map over all hard-interference SNR–SIR cells. Each entry is A5 minus the named strong baseline in macro-F1 percentage points; a star marks a positive unadjusted paired lower interval, while Holm-surviving cells are reported separately in the text. Displaying the full map prevents the severe corner from being read as a uniform ranking.

analysis into confirmation, and the severe corner is an in-sample stratification of the hard-interference cells, not a held-out validation.

C. Empirical Occupancy–SIR Operating Envelope

The rank exchange motivates an empirical law rather than another marginal average. For reference b, we fit

$$\Delta_b = \beta_{0,b} + \beta_{1,b} o + \beta_{2,b}\,\mathrm{SIR}, \tag{8}$$

where o is jammer spectral occupancy and Δ_b is A5 minus the reference macro-F1. Occupancy is defined as the fraction of time–frequency cells whose tracked jammer power exceeds a noise-floor-referenced threshold, aggregated over the STFT lattice of a window and then over the cells of an evaluation split:

$$o = \frac{1}{FT} \sum_{f,t} \mathbf{1}\left(P_j(f,t) > \tau\right). \tag{9}$$

Only the 32 in-distribution and hard-interference cells enter the fit; the remaining 81 cells (unseen jammer, unseen speed, held-out channel, combined OOD, and clean retention) are held out. Occupancy and SIR are nearly orthogonal in this design (pooled Pearson $r = -0.03$), so Eq. (8) is a genuine two-variable fit rather than a confounded one; it is nevertheless treated as descriptive rather than causal because occupancy is simulator-derived and the design is observational with respect to this covariate.

Against IQFormer-inspired, the fitted occupancy coefficient is 11.39 percentage points per unit occupancy and the SIR coefficient is -0.666 percentage points per dB. On the 81 held-out cells the envelope attains RMSE 6.20 percentage points, versus 15.28 percentage points for a constant predictor that outputs the mean of the fitting cells, for $R^2_{\mathrm{skill}} = 0.84$ (Fig. 4). Sign agreement is 0.975, but 0.963 of held-out gains are negative, so a majority-sign predictor already attains 0.963; sign accuracy is therefore reported for completeness and is not the principal validation. The MCLDNN envelope gives RMSE 8.05 percentage points versus a constant baseline of 13.29 percentage points ($R^2_{\mathrm{skill}} = 0.63$) and sign agreement 0.951 against a majority-sign baseline of 0.901.

TABLE VI
EXPLORATORY BREAK-EVEN OCCUPANCY UNDER SIMULATOR-TRACKED JAMMER OCCUPANCY.

SIR (dB)	vs IQFormer-inspired	vs MCLDNN
-15	0.58	0.48
-10	0.87	0.66

Solving Eq. (8) for zero gain yields the break-even occupancies in Table VI. Because occupancy is a simulator-tracked quantity, these are simulator-domain reference values conditional on an occupancy estimate, not deployment-ready control thresholds. The required occupancy rises as interference becomes less severe.

The preregistered mechanism expectation was non-increasing gain with occupancy. The sealed directional test rejected that sign. Exploratory quintiles instead show monotonically increasing A5–CSSL gain, from a negative low-occupancy region to a strongly positive high-occupancy region. We return to the interpretation of this reversal in Section VI-C.

D. Complexity, Complementarity, and Selective Prediction

A5 remains useful outside its winning region because its errors differ from those of the I/Q models. Probability fusion with IQFormer-inspired improves over the stronger constituent by 4.71 percentage points, whereas a same-architecture seed-ensemble control improves by only 1.49 percentage points; the corresponding MCLDNN fusion and control gains are 5.70 and 1.69 percentage points. This is supporting evidence of complementary decisions, not evidence that the compact model alone wins the marginal benchmark.

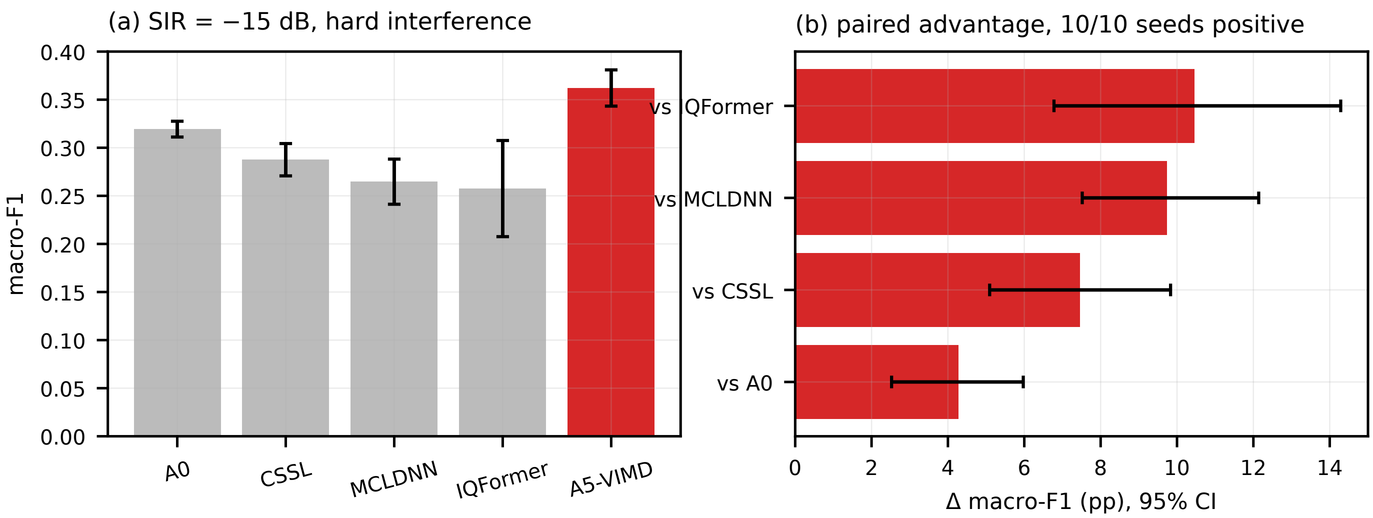

Fig. 3. Severe-interference corner. Left: absolute macro-F1 at SIR $= -15$ dB. Right: source-paired A5 advantages with stratified bootstrap intervals. The comparison is exploratory and post hoc.

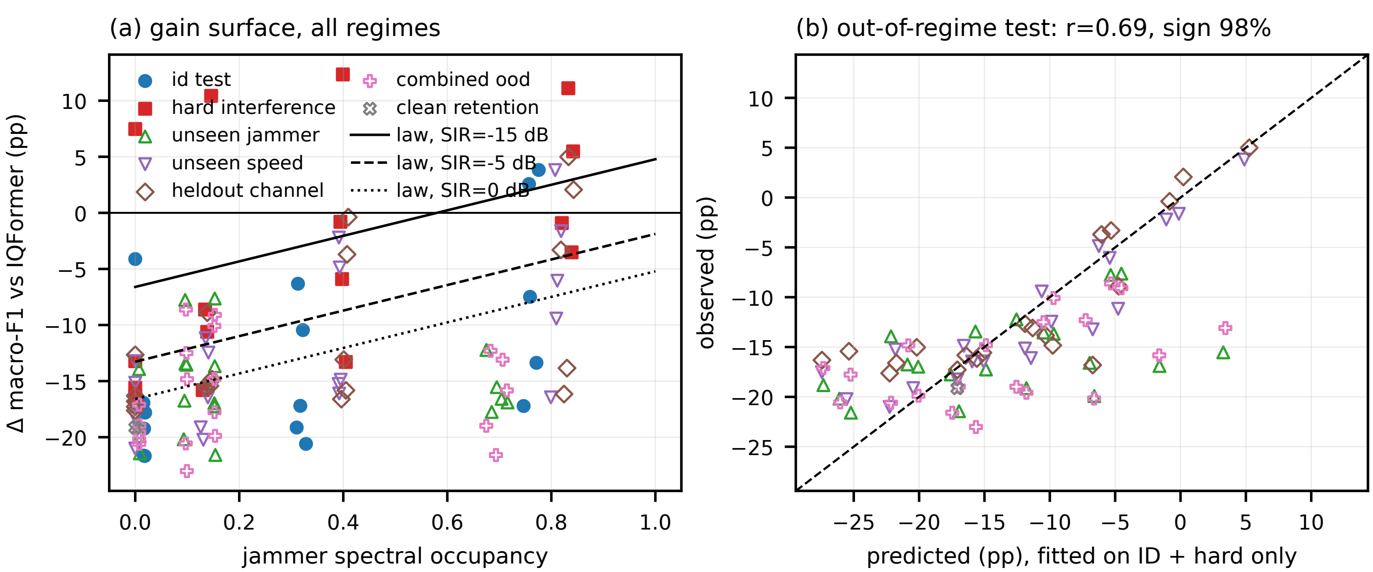

Fig. 4. Exploratory occupancy–SIR envelope. Filled cells fit the law; open cells are held-out regimes. The right panel compares out-of-regime predictions with artifact-derived outcomes; the caption reports RMSE relative to a constant baseline and R^2_{skill}, not sign accuracy alone.

A5 uses 39500 parameters, 41.8 million MACs, and a median latency of 3.172 ms, versus 354984 parameters, 355.6 million MACs, and 4.180 ms for IQFormer-inspired, and 406070 parameters, 398.2 million MACs, and 3.789 ms for MCLDNN. The compact advantage is thus strongest in parameter count and MACs; the median-latency advantage over MCLDNN is modest, so we frame compactness as a parameter and MACs property rather than an end-to-end inference-cost claim. At 30% retained coverage after validation-only confidence calibration, A5 achieves 94.4% selective accuracy, close to the IQFormer-inspired 94.9% (Fig. 6). Selective prediction is supporting rather than central evidence.

E. Clean-Boundary Diagnosis

The clean-retention split contains all ten modulations, but the two training views expose jammer-free windows for only 4 classes; the remaining 6 classes are a condition-transfer test. Among uncovered constellation-order cells, transfer failure occurs in 100% of compact-spectral cells versus 11% of I/Q high-capacity cells. The clean-retention gate did not pass. Figure 7 localizes the failure rather than hiding it in an aggregate.

We then followed a prospective decision chain (Table VII). Frozen-checkpoint linear and MLP probes used a correctly reconstructed jammer-free counterfactual and source-disjoint clean cross-validation. For A5, 5 of 6 design–seed cells lie below the preregistered high-order threshold 0.25; one cell is borderline. The strong I/Q baselines are above that threshold in 18 of 18 cells. The evidence therefore leans toward missing constellation information in the compact spectral representation rather than a classification head that fails to use available information.

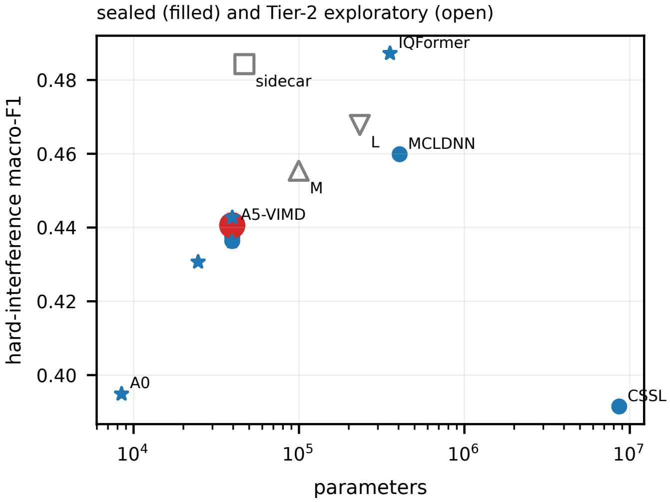

Fig. 5. Hard-interference macro-F1 versus parameter count. Sealed models are solid; Tier-2 sidecar and capacity tiers are open and labeled exploratory. The frontier is regime-specific.

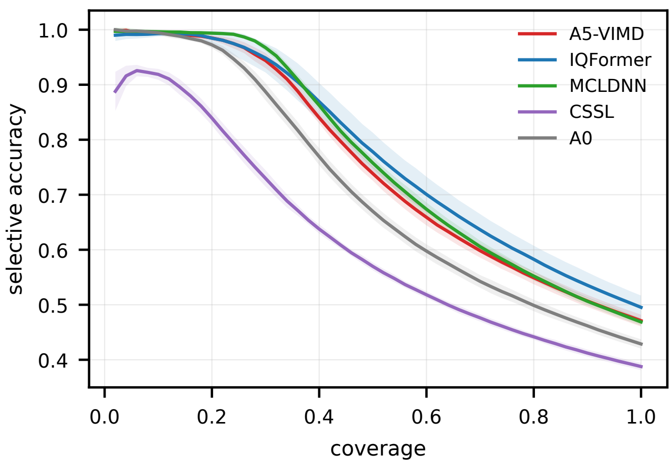

Fig. 6. Selective accuracy versus coverage on hard interference. Thresholds are calibrated without using test labels. This is supporting evidence.

Adding clean windows for every class tests the simpler coverage explanation. It changes clean macro-F1 by only 0.15 percentage points($[-0.31, 0.63]$) and hard macro-F1 by -0.30 percentage points ($[-0.91, 0.27]$). Coverage alone is therefore insufficient. A preregistered negative control gates the modulation route using predicted interference presence. Relative to A5, it changes clean macro-F1 by -0.30 percentage points($[-0.66, 0.08]$) and hard macro-F1 by -0.18 percentage points($[-0.72, 0.35]$). The learned gate averages 99.994% on clean windows and 99.991% under hard interference, so it saturates near the modulated route in both conditions rather than implementing the intended clean identity fallback. This arm is a negative control, not stand-alone evidence that a functional mask fallback was learned.

F. Prospective I/Q Repair and Capacity Controls

The next intervention adds a received-I/Q sidecar while retaining the spectral routes and the same source mixture. It brings the total model size to 46794 parameters (7294 parameters beyond A5, +18%) and 43.1 million MACs (about +3%). It raises clean macro-F1 by 11.77 percentage points($[11.06, 12.49]$) and hard macro-F1 by 4.37 percentage points($[3.44, 5.31]$). Its overall hard score is statistically near IQFormer-inspired ($\Delta = -0.29$ percentage points, $[-1.95, 1.46]$), while at severe SIR it remains ahead by 12.09 percentage points($[8.32, 15.88]$). The sidecar was designed prospectively after the clean-boundary diagnosis; it is interpreted as an intervention on the hypothesized information bottleneck, not as retroactive evidence for the original A5 architecture and not as a new multimodal fusion architecture. This repairs the clean boundary without erasing the structured-interference region.

Finally, we widen the spectral model without changing its representation domain. The S, M, and L tiers contain 39500, 99596, and 233244 parameters. Relative to S, M improves hard macro-F1 by 1.94 percentage points and L by 3.18 percentage points; L improves over M by 1.24 percentage points. Capacity therefore improves average fitting. The capacity ladder uses a five-seed matched subset for both candidates and references, so its comparator means differ slightly from the ten-seed main results; Table VIII uses the matched subset throughout. L remains below IQFormer-inspired on overall hard interference by -2.84 percentage points($[-4.30, -1.31]$). At SIR $= -15$ dB, the same L model is ahead by 14.07 percentage points($[9.55, 18.51]$). Increasing spectral capacity therefore shifts performance but does not remove the operating boundary, supporting a representation interpretation rather than a pure parameter-count explanation. Note that the S tier and A5 share the 39.5k-parameter budget but differ in configuration details, so their hard-interference scores are not identical.

VI. Discussion

A. What the Operating Envelope Establishes

The evidence supports a practical model-selection statement within the simulator domain. A compact spectral front end is most attractive when interference is both severe and occupies enough of the time–frequency plane for learned allocation to matter. A larger I/Q model is preferable in weak, clean, or low-occupancy conditions unless the compact system includes an I/Q side channel. The two models are also complementary enough that late probability fusion can outperform either alone. Occupancy and SIR are nearly orthogonal in this design, so the two-variable envelope is not a confounded one-dimensional fit.

B. What It Does Not Establish

Eq. (8) is empirical and descriptive, not a causal structural equation. Jammer occupancy is estimated from simulator-tracked components and is not directly observable at an uncooperative receiver; the break-even occupancies of Table VI are therefore simulator-domain reference values conditional on an occupancy estimate. The severe SIR corner is an in-sample stratification, and the held-out evaluation covers regimes within the same frozen simulator campaign, not real-world out-of-distribution conditions.

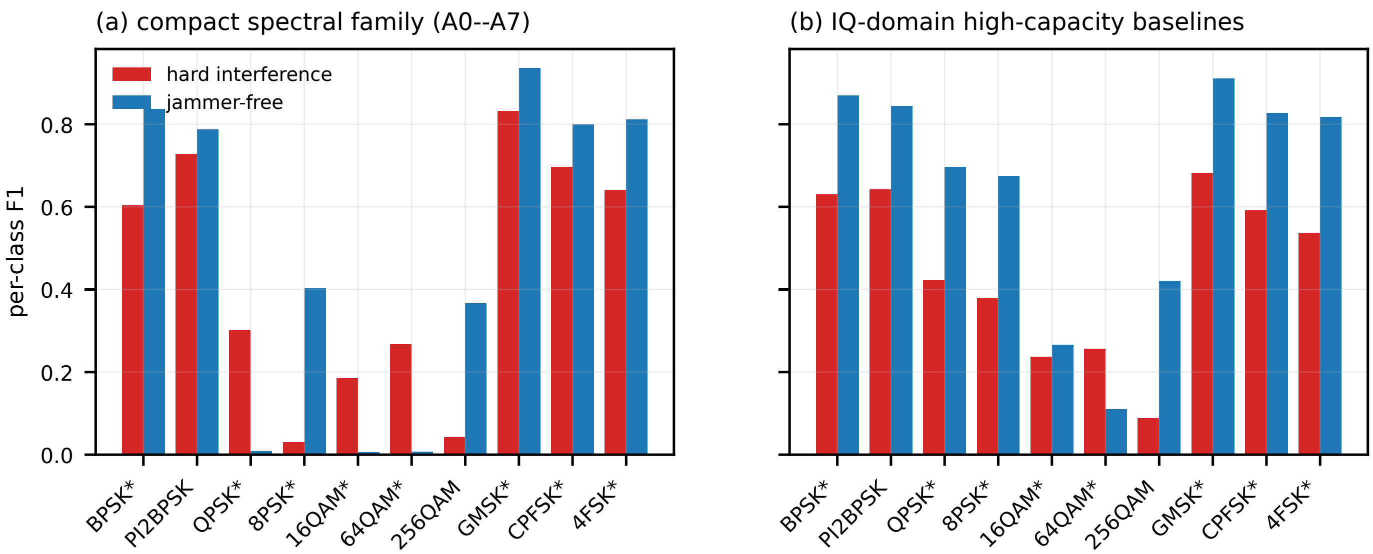

Fig. 7. Condition-transfer taxonomy. A star marks a class whose jammer-free condition is uncovered during training; training coverage and discriminability type separate the compact spectral failure from classes whose envelope cues remain distinctive.

TABLE VII
PROSPECTIVE TIER-2 DIAGNOSIS, REPAIR, AND NEGATIVE CONTROL. THESE RESULTS ARE EXPLORATORY AND OUTSIDE THE FROZEN CONFIRMATORY FAMILY.

Intervention	Question	Params	MACs	Clean diff. [95% CI]	Hard diff. [95% CI]	Decision
Per-class clean coverage	Is exposure alone sufficient?	39.5k	41.8M	0.15 [-0.31,0.63]	-0.30 [-0.91,0.27]	No
Presence-gated route vs A5	Is mask application the primary clean cause?	39.5k	41.8M	-0.30 [-0.66,0.08]	-0.18 [-0.72,0.35]	No; $g \approx 1$ in both conditions
Received-I/Q sidecar vs A5	Is the representation gap repairable?	46.8k	43.1M	11.77 [11.06,12.49]	4.37 [3.44,5.31]	Yes; both intervals positive

TABLE VIII
EXPLORATORY CAPACITY LADDER ON HARD INTERFERENCE (FIVE-SEED MATCHED SUBSET).

Contrast	Diff.	Low	High
M minus S	1.94	1.07	2.80
L minus S	3.18	2.44	3.96
L minus M	1.24	0.31	2.37
L minus IQFormer-inspired	-2.84	-4.30	-1.31

C. Meaning of the Failed Directional Hypothesis

The preregistered directional mechanism hypothesis was not supported. The data reject the expectation that the compact model's relative gain should decrease with increasing occupancy. The surviving result is therefore empirical rather than mechanistic: occupancy organizes the observed rank exchange, but the observed direction should not be interpreted as confirmation of the originally hypothesized masking mechanism. In this paper, "physics-informed" refers to the construction of the training supervision, not to a confirmed monotonic physical law.

D. Representation Repair and Practical Model Selection

The sequence of interventions matters. The frozen probe narrows the cause; the coverage arm rejects a cheap explanation; the I/Q sidecar repairs the predicted missing information; and the capacity ladder shows that width alone does not eliminate the boundary. The Tier-2 evidence is more consistent with a representation-information gap than with insufficient clean exposure alone. The sidecar is a prospective intervention on that hypothesis, not the new main classifier and not a multimodal-fusion novelty.

E. External Validity and Deployment Proxies

A future prospective campaign should randomize occupancy and SIR more independently and estimate a receiver-observable occupancy proxy (for example occupied-bandwidth or spectral-kurtosis estimators). The present campaign establishes an internally controlled operating boundary; validation with receiver-observable occupancy estimates and over-the-air or public interference recordings is a separate external-validity step.

VII. LIMITATIONS AND REPRODUCIBILITY

The evidence has the following explicit limits. First, all results are from source-disjoint TR 38.901 TDL-profile simulations, not a complete vehicular geometry or a MAC-/network-layer V2X evaluation. Second, the teacher uses simulation-component bookkeeping and cannot supervise unreferenced recordings. Third, occupancy is simulator-derived and is not directly available at a receiver. Fourth, the original confirmatory family gate did not pass; only the complete A5–

A0 contrast is a positive sealed result. Fifth, the clean-retention gate did not pass. Sixth, the severe-interference, operating-envelope, fusion, condition-transfer, probe, sidecar, and capacity analyses are exploratory or post-hoc, with post-hoc status stated where applicable. Seventh, public or over-the-air external validation is not yet established. Eighth, the IQFormer-inspired and MCLDNN comparators are unified-retraining references, not exact reproductions of every original training recipe.

Reproducibility is enforced at the artifact boundary. Each manuscript macro stores the source path, row selector, evidence class, precision, and source SHA-256; the resolved macro appendix in this submission exposes the numerical values and evidence class, and the full provenance manifest is provided with the reproduction package. Figures are rendered from CSV files and carry a separate provenance manifest. The frozen run stores cache identity, configurations, checkpoints, source-aligned predictions, seed aggregates, paired statistics, environment metadata, and execution records. These controls prevent a favorable Tier-2 result from overwriting an unfavorable sealed gate.

VIII. Conclusion

Compact spectral and high-capacity I/Q representations exchange rank across the structured-interference plane in a reproducible, condition-dependent way. An occupancy–SIR envelope describes that exchange with meaningful magnitude skill over held-out regime cells relative to a constant baseline, and a post-hoc but statistically reinforced severe corner shows a reproducible advantage for the compact model. The sealed whole-configuration contrast over a shared backbone is positive, while two isolated component effects are bounded and the confirmatory family gate does not pass; the clean-retention gate also does not pass, but a probe–coverage–sidecar chain localizes and repairs the representation gap, and a capacity ladder shows that parameter count alone cannot explain the remaining boundary. The result is an honest simulation-only operating envelope with a concrete repair path, not a claim of uniform superiority.

References

[1] T. J. O'Shea, J. Corgan, and T. C. Clancy, "Convolutional radio modulation recognition networks," in *Engineering Applications of Neural Networks*, 2016, pp. 213–226.

[2] F. Meng, P. Chen, L. Wu, and X. Wang, "Automatic modulation classification: A deep learning enabled approach," *IEEE Trans. Veh. Technol.*, vol. 67, no. 11, pp. 10 760–10 772, 2018.

[3] Y. Tu, Y. Lin, C. Hou, and S. Mao, "Complex-valued networks for automatic modulation classification," *IEEE Trans. Veh. Technol.*, vol. 69, no. 9, pp. 10 085–10 089, 2020.

[4] Z. Zhang, H. Luo, C. Wang, C. Gan, and Y. Xiang, "Automatic modulation classification using CNN-LSTM based dual-stream structure," *IEEE Trans. Veh. Technol.*, vol. 69, no. 11, pp. 13 521–13 531, 2020.

[5] J. Xu, C. Luo, G. Parr, and Y. Luo, "A spatiotemporal multi-channel learning framework for automatic modulation recognition," *IEEE Wireless Commun. Lett.*, vol. 9, no. 10, pp. 1629–1632, 2020.

[6] J. Cai, F. Gan, X. Cao, and W. Liu, "Signal modulation classification based on the transformer network," *IEEE Trans. Cogn. Commun. Netw.*, vol. 8, no. 3, pp. 1348–1357, 2022.

[7] M. Shao, D. Li, S. Hong, J. Qi, and H. Sun, "IQFormer: A novel transformer-based model with multi-modality fusion for automatic modulation recognition," *IEEE Trans. Cogn. Commun. Netw.*, vol. 11, no. 3, pp. 1623–1634, 2025.

[8] Y. Wang, J. Yang, M. Liu, and G. Gui, "LightAMC: Lightweight automatic modulation classification via deep learning and compressive sensing," *IEEE Trans. Veh. Technol.*, vol. 69, no. 3, pp. 3491–3495, 2020.

[9] S. Sharma *et al.*, "Efficient automatic modulation classification for next-generation wireless networks," *IEEE Trans. Green Commun. Netw.*, 2026, author list to be verified before submission.

[10] S. Wang, H. Xing, C. Wang, H. Zhou, B. Hou, and L. Jiao, "SigDA: A superimposed domain adaptation framework for automatic modulation classification," *IEEE Trans. Wireless Commun.*, vol. 23, no. 10, pp. 13 159–13 172, 2024.

[11] C. Xiao, S. Yang, Z. Feng, and L. Jiao, "MCLHN: Toward automatic modulation classification via masked contrastive learning with hard negatives," *IEEE Trans. Wireless Commun.*, vol. 23, no. 10, pp. 14 304–14 319, 2024.

[12] M. Du, J. Pan, and D. Bi, "A contrastive learner for automatic modulation classification," *IEEE Trans. Wireless Commun.*, vol. 24, no. 4, pp. 3575–3589, 2025.

[13] X. Sun, C. Wu, Y. Li, Y. Li, T. Huang, J. Yu, and H. Zhang, "Toward interference-tolerant automatic modulation recognition via multi-stage feature extraction network," *IEEE Trans. Veh. Technol.*, vol. 74, no. 8, pp. 12 629–12 640, 2025.

[14] G. Jajoo and P. Singh, "Modulation classification for overlapped signals using deep learning," *IEEE Open J. Commun. Soc.*, vol. 5, pp. 3839–3851, 2024.

[15] A. Faysal, M. Rostami, T. Boushine, R. G. Roshan, N. Muralidhar, Y.-D. Yao, and H. Wang, "DenoMAE2.0: Improving denoising masked autoencoders by classifying local patches for automatic modulation classification," *IEEE Trans. Commun.*, vol. 74, pp. 929–943, 2026.

[16] W. Li, Z. Chen, T. Liu, H. Chen, and G. Xu, "Confidence-guided prototypical contrastive domain adaptation for cross-domain automatic modulation classification," *IEEE Trans. Cogn. Commun. Netw.*, vol. 12, pp. 6929–6941, 2026.

[17] J. Zhang, X. Zhang, W. X. Zheng, Y. Chen, N. Zhao, J. Li, and M. Liu, "Modulation recognition via tensor analysis for MIMO systems with malicious interference," *IEEE Trans. Veh. Technol.*, vol. 75, no. 8, pp. 18 609–18 613, 2026.

[18] "Dual knowledge and data-driven network for cross-domain automatic modulation classification (DKDNet)," arXiv:2607.08031, 2026, preprint; author list to be verified.

[19] "MoEFormer for automatic modulation classification," arXiv:2606.09085, 2026, preprint; full title and author list to be verified.

[20] "ZoomSpec: Physics-guided coarse-to-fine spectrum sensing with time–frequency fusion," arXiv:2604.13568, 2026, preprint; full title and author list to be verified.

[21] "RIS-MAE: Masked autoencoding for automatic modulation classification," arXiv:2508.00274, 2025, preprint; full title and author list to be verified.

[22] M. Du, J. Pan, and D. Bi, "CSSL-AMC-Pytorch source-code snapshot," GitHub repository, commit 2fbc5b3e12f780b0b26eb0ee2c33d592739aa24f, https://github.com/dumingyang20/CSSL-AMC-Pytorch/commit/2fbc5b3e12f780b0b26eb0ee2c33d592739aa24f, 2025, apache-2.0 license; accessed Jul. 28, 2026.